

Comprehensive Analysis of Banking Customer and Transaction Data to Identify Financial Trends, Customer Behavior, and Business Insights

1. Project Overview

This project focuses on analyzing banking customer data to understand customer behavior, financial performance, credit risk, and transaction patterns. Using Python and Power BI, raw banking data was cleaned, transformed, analyzed, and visualized to generate meaningful business insights and support data-driven decision-making.

Project Objective

- To analyze customer demographics and banking behavior
- To identify customer segments and financial trends
- To monitor account balances and annual income patterns
- To evaluate customer credit scores and risk ratings
- To analyze transaction channels and digital banking adoption
- To create interactive dashboards for business insights

2. Data Source

Source Description and Timeline: [Banking Sector Anonymized Dataset](#)

Domain: Banking & Financial Services

3. Problem Statement

Analyzing banking customer data by cleaning, transforming, and visualizing datasets using Python and Power BI to identify customer behavior patterns, account balance trends, credit risk profiles, transaction channel usage, and digital banking adoption. The project aims to generate actionable insights that support business growth and customer-focused decision-making.

4. Attribute (Column /Features) Details: Attribute

Attribute Name	Data Type	Description
customer_id	String (Text)	Unique identification number of the customer
account_id	String (Text)	Unique identification number of the bank account
account_type	String (Text)	Type of bank account (Savings, Current, Salary, etc.)

account_open_date	Date	Date when the account was opened
account_status	String (Text)	Current status of the account (Active, Inactive, Closed)
gender	String (Text)	Gender of the customer
age	Integer (Numeric)	Age of the customer
state	String (Text)	State where the customer belongs
pincode	Integer (Numeric)	Area postal code of the customer
customer_segment	String (Text)	Category of customer based on banking profile
employment_status	String (Text)	Employment status of the customer
employer_name	String (Text)	Name of the customer's employer/company
annual_income_inr	Decimal (Numeric)	Annual income of the customer in INR
account_balance_inr	Decimal (Numeric)	Current account balance in INR
num_products_held	Integer (Numeric)	Number of banking products used by the customer
kyc_status	String (Text)	Status of KYC verification
nominee_added	String (Text)	Indicates whether nominee is added or not
mobile_banking_enabled	String (Text)	Indicates whether mobile banking is enabled
net_banking_enabled	String (Text)	Indicates whether net banking is enabled
credit_card_linked	String (Text)	Indicates whether a credit card is linked
demat_account	String (Text)	Indicates whether the customer has a demat account
branch_name	String (Text)	Name of the bank branch
relationship_manager_id	String (Text)	ID of the assigned relationship manager
last_login_days_ago	Integer (Numeric)	Number of days since last login
is_dormant	Integer (Numeric)	Indicates whether the account is dormant
churn_flag	Integer (Numeric)	Indicates whether the customer left the bank
fraud_flag	Integer (Numeric)	Indicates suspected fraudulent activity
loan_type	String (Text)	Type of loan taken by the customer

loan_amount_inr	Decimal (Numeric)	Total loan amount in INR
loan_tenure_months	Integer (Numeric)	Duration of the loan in months
outstanding_principal_inr	Decimal (Numeric)	Remaining principal loan amount
emi_amount_inr	Decimal (Numeric)	Monthly EMI payment amount
days_past_due	Integer (Numeric)	Number of overdue payment days
is_npa	Integer (Numeric)	Indicates whether the loan is a Non-Performing Asset
penalty_amount_inr	Decimal (Numeric)	Penalty amount charged for overdue payments
credit_score	Integer (Numeric)	Customer credit score
credit_bureau	String (Text)	Credit bureau source information
risk_rating	String (Text)	Risk category assigned to the customer
savings_interest_rate_pct	Decimal (Numeric)	Savings account interest rate percentage
fd_interest_rate_pct	Decimal (Numeric)	Fixed deposit interest rate percentage
overdraft_limit_inr	Decimal (Numeric)	Overdraft limit provided to the customer
monthly_interest_income_inr	Decimal (Numeric)	Monthly interest income earned by the bank
monthly_service_charges_inr	Decimal (Numeric)	Monthly service charges collected
num_transactions_last_month	Integer (Numeric)	Total transactions made in the last month
avg_transaction_value_inr	Decimal (Numeric)	Average transaction value in INR
last_transaction_date	Date	Date of the latest transaction
last_transaction_type	String (Text)	Type of latest transaction
last_transaction_channel	String (Text)	Channel used for the latest transaction
last_transaction_amount_inr	Decimal (Numeric)	Amount of the latest transaction
transaction_currency	String (Text)	Currency used in the transaction

5. Tools & Technologies

Python

- Data Cleaning
- Data Transformation

- Exploratory Data Analysis (EDA)
- Data Validation

Power BI

- Data Modelling
- DAX Calculations
- Dashboard Development
- Interactive Visualizations

6. Data Preprocessing

Data Cleaning Activities

- Removed duplicate records
- Handled missing values
- Corrected data types
- Standardized categorical values
- Renamed columns for consistency
- Verified data quality and accuracy

7. Screenshot of Python Data Cleaning and Transformation

Problem Definition and Dataset Selection

```
# Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Load dataset
df = pd.read_csv("https://raw.githubusercontent.com/arthi-rathinam/Main-Project/refs/heads/main/Banking%20Sector%20Anonymised%20Dataset%20-%20Raw%20-uncleaned.csv")
```

```
# Dataset description (rows, columns, features) - Data Shape
print("\n Dataset Shape : ", df.shape)
print("\n Number of Rows : ", df.shape[0])
print("\n Number of Columns : ", df.shape[1])
```

```
...
Dataset Shape : (500, 50)

Number of Rows : 500

Number of Columns : 50
```

```
# view Dataset columns
print("\n Features/Columns : ", "\n", df.columns)

... Show hidden output

df.size

25000

# DataSet information
print("\n Dataset Information : ")
df.info()

Show hidden output

# statistical summary
print("\n Statistical Summary : ")
df.describe()

Show hidden output
```

```
# Initial EDA (head, info, describe, shape, null checks, duplicate check)
# View First 5 Rows
df.head()
```

	customer_id	account_id	account_type	account_open_date	account_status
0	CUST100000	ACC900000	Savings	13-04-2018	Inactive
1	CUST100001	ACC900001	Current	18-08-2018	NaN
2	CUST100002	ACC900002	Recurring Deposit	20-08-2021	Active
3	CUST100003	ACC900003	Checking	24-02-2020	Closed
4	CUST100004	ACC900004	Fixed Deposit	19-12-2024	Inactive

5 rows x 50 columns

```
# Bottom First 5 Rows
df.tail()
```

	customer_id	account_id	account_type	account_open_date
495	CUST100495	ACC900495	Current	07-10-2022
496	CUST100496	ACC900496	Checking	29-10-2024
497	CUST100497	ACC900497	Savings	11-09-2021
498	CUST100498	ACC900498	Current	07-06-2019
499	CUST100499	ACC900499	Fixed Deposit	25-04-2021

5 rows x 50 columns

```
# Check Null Values
df.isnull().sum()
```

	gender	age	state	pincode	customer_segment	employment_status	employer_name	annual_income_inr	account_balance_inr	num_products_held	kyc_status	nominee_added
...	98	23	40	16	98	96	427	17	18	9	109	145

```
# Check duplicate Records
df.duplicated().sum()

... np.int64(0)
```

```
# Datatype
df.dtypes
```

customer_id	object	
account_id	object	
account_type	object	
account_open_date	object	
account_status	object	
gender	object	
age	float64	

```
# Check Null values
df.nunique()
```

customer_id	500	
account_id	500	
account_type	5	
account_open_date	468	
account_status	4	
gender	3	
age	58	
state	8	
pincode	483	
customer_segment	4	

```
# Display random sample data
df.sample(4)
```

	customer_id	account_id	account_type	account_open_date	account_status
198	CUST100198	ACC900198	Checking	07-05-2020	Active
474	CUST100474	ACC900474	Checking	16-06-2024	Closed
304	CUST100304	ACC900304	Savings	03-09-2023	Inactive
194	CUST100194	ACC900194	Savings	24-01-2019	Active

4 rows x 50 columns

Data Cleaning and Pre-processing

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Load dataset
df = pd.read_csv("https://raw.githubusercontent.com/arthi-rathinam/Main-Project/refs/heads/main/Banking%20Sector%20Anonymised%20Dataset%20-%20Raw%20-uncleaned.csv")
```

```
# Handling duplicate values
df.duplicated().sum()

np.int64(0)
```

```
# Handling missing values
df.isnull().sum()
```

customer_id	0	
account_id	0	
account_type	0	
account_open_date	0	
account_status	105	
gender	98	

```
# Dropped the columns
print("\n Drop the Columns : ",df.drop(columns=['pincode', 'employer_name', 'relationship_manager_id','savings_interest_rate_pct', 'fd_interest_rate_pct',
                                                'overdraft_limit_inr','num_products_held','branch_name','credit_bureau'], inplace=True))
df.head(5)
```

```
# Handling missing values
df['age'] = df['age'].fillna(df['age'].median())
```

```
# Handling missing values
df['account_status'] = df['account_status'].fillna(df['account_status'].mode()[0])
df['gender'] = df['gender'].fillna(df['gender'].mode()[0])
```

```
# Handling missing values - for Categorical col
cols = ['state', 'customer_segment', 'employment_status', 'kyc_status', 'nominee_added', 'mobile_banking_enabled',
        'net_banking_enabled', 'credit_card_linked', 'demat_account', 'risk_rating']

for col in cols:
    df[col] = df[col].fillna(df[col].mode()[0])
```

```
df['annual_income_inr'] = df['annual_income_inr'].fillna(df['annual_income_inr'].median())

df['account_balance_inr'] = df['account_balance_inr'].fillna(df['account_balance_inr'].median())

df['last_login_days_ago'] = df['last_login_days_ago'].fillna(df['last_login_days_ago'].median())
```

```
binary_columns = ['is_dormant', 'churn_flag', 'fraud_flag','is_npa']
```

```
for col in binary_columns:
    df[col] = df[col].fillna(df[col].mode()[0])
```

```
df['credit_score'] = df['credit_score'].fillna(df['credit_score'].mean())

df['num_transactions_last_month'] = df['num_transactions_last_month'].fillna(df['num_transactions_last_month'].mean())
```

```
df['loan_type'] = df['loan_type'].fillna('No Loan')

df['loan_amount_inr'] = df['loan_amount_inr'].fillna(0)

df['loan_tenure_months'] = df['loan_tenure_months'].fillna(0)

df['outstanding_principal_inr'] = df['outstanding_principal_inr'].fillna(0)

df['emi_amount_inr'] = df['emi_amount_inr'].fillna(0)
```

```
df.loc[df['loan_type'] == 'No Loan', 'days_past_due'] = 0
df['days_past_due'] = df['days_past_due'].fillna(df['days_past_due'].mean())
```

```
df.loc[df['loan_type'] == 'No Loan', 'penalty_amount_inr'] = 0
df['penalty_amount_inr'] = df['penalty_amount_inr'].fillna(df['penalty_amount_inr'].median())
```

```
df['monthly_interest_income_inr'] = df['monthly_interest_income_inr'].fillna(df['monthly_interest_income_inr'].median())
df['monthly_service_charges_inr'] = df['monthly_service_charges_inr'].fillna(df['monthly_service_charges_inr'].median())
df['avg_transaction_value_inr'] = df['avg_transaction_value_inr'].fillna(df['avg_transaction_value_inr'].median())
df['last_transaction_amount_inr'] = df['last_transaction_amount_inr'].fillna(df['last_transaction_amount_inr'].median())
```

```
df['gender'] = df['gender'].replace({'M' : 'Male', 'F' : 'Female'})
```

```
df['account_open_date'] = pd.to_datetime(df['account_open_date'],format='%d-%m-%Y')
df['account_open_date'] = df['account_open_date'].dt.strftime('%d/%m/%Y')

df['last_transaction_date'] = pd.to_datetime(df['last_transaction_date'],format='%d-%m-%Y')
df['last_transaction_date'] = df['last_transaction_date'].dt.strftime('%d/%m/%Y')
```

```
df.columns = df.columns.str.strip().str.capitalize()
```

```
df.columns = df.columns.str.replace('inr', 'INR')
```

```
df = df.drop_duplicates()
```

```
df.isnull().sum()
```

```
0
Customer_id    0
Account_id     0
Account_type   0
Account_open_date  0
Account_status  0
Gender         0
```

```
df.isnull().sum().sum()
```

```
np.int64(0)
```

```
df.shape
```

```
(500, 41)
```

Steps to Download your cleaned data for dashboard creation using Power BI

```
df.to_csv("Banking_Cleaned_Data.csv", index=False)
print("File saved successfully")
... File saved successfully
```

```
import os
print(os.listdir())

from google.colab import files
files.download("Banking_Cleaned_Data.csv")

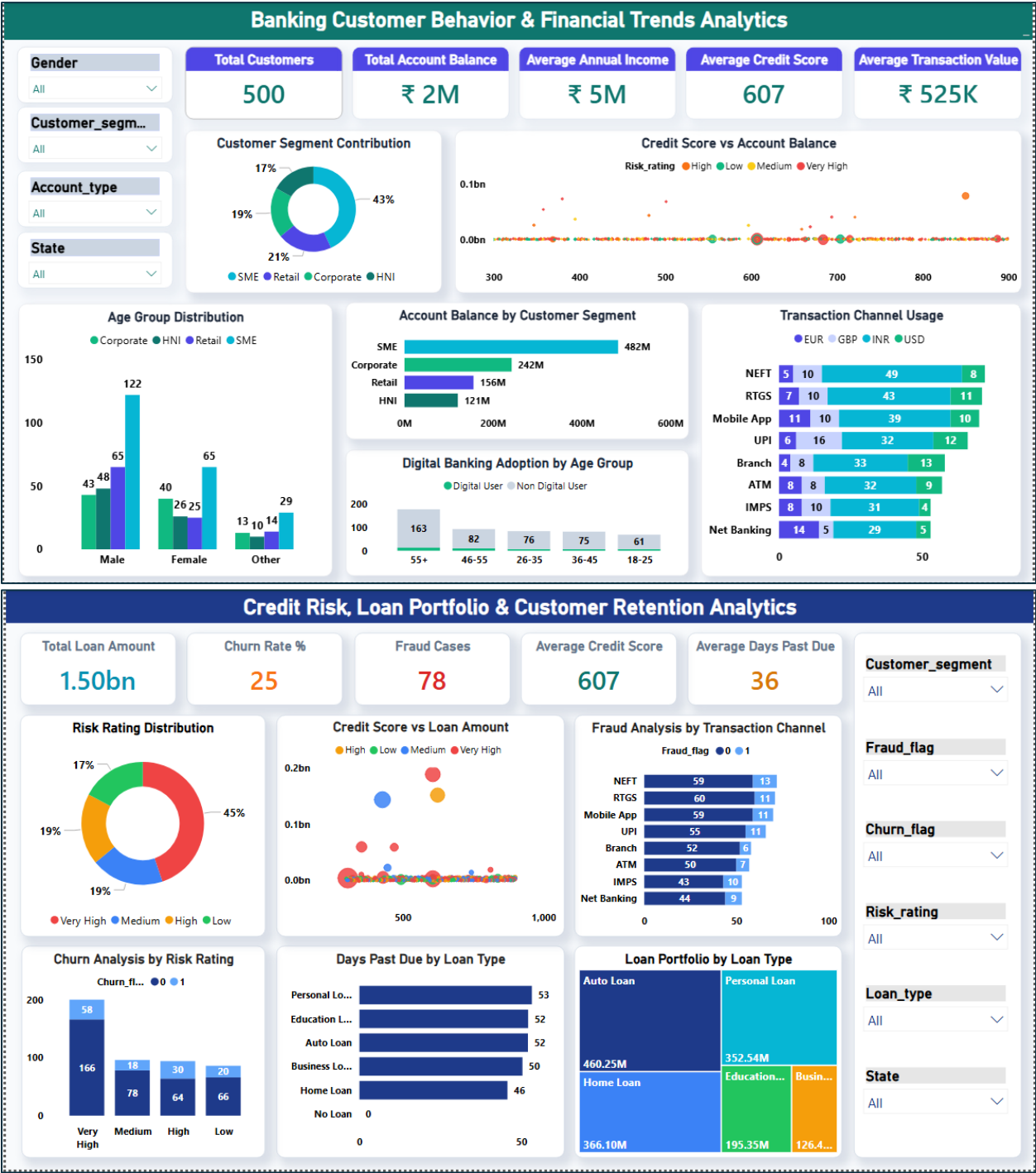
['.config', 'Banking_Cleaned_Data.csv', 'sample_data']
```

10. Analysis and Visualizations (Power BI) Dashboard Features

The Power BI dashboard includes KPI Cards, Bar Charts, Line Charts, Pie Charts, Donut Charts, Funnel Charts, Column Charts, Scatter Charts, Tree map Charts, Gauge Charts, Tables, Matrix Tables, Slicers, and Interactive Filters to communicate customer behavior, account performance, loan portfolio analysis, credit risk assessment, digital banking adoption, transaction patterns,

fraud monitoring, churn analysis, and overall banking performance insights.

Screenshot of Dashboard



11. Insights & Conclusions

Key Findings:

- SME customers constitute the largest customer segment in the dataset, indicating their significant contribution to the bank's customer base and overall business activity
- Corporate customers maintain the highest average account balances, reflecting larger financial operations and transaction volumes
- Digital banking adoption is high among customers, demonstrating a strong preference for technology-driven banking services
- NEFT, RTGS, and Mobile Banking are the most frequently used transaction channels, highlighting the growing importance of digital transactions
- Most customers fall within the moderate credit score category, suggesting a balanced credit risk profile across the customer base
- Customers with higher annual incomes generally maintain higher account balances, indicating a positive relationship between income and banking activity
- Transaction patterns vary across customer segments, showing different banking needs and service usage behaviors
- The increasing use of digital channels suggests that future banking growth will be driven by mobile and online banking services
-

12. Conclusion

The banking customer data analysis successfully identified key patterns in customer demographics, account balances, credit scores, and transaction behavior. The findings highlight the importance of SME customers, digital banking adoption, and customer segmentation in driving banking performance. These insights can support strategic decision-making, improve customer experience, and help the bank develop targeted products and services.